

BALTIC STOCK MARKETS DISTRIBUTIONAL-SHAPE HOMOGENITY – A WASSERSTEIN DISTANCE ANALYSIS

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ABSTRACT. This project aims to investigate whether Baltic stock markets exhibit homogeneous dynamics in the shape of their return distributions. The analysis applies the 2-Wasserstein distance to rolling normalized empirical distributions of daily returns for the Tallinn (OMXT), Riga (OMXR), and Vilnius (OMXV) stock indices, with the DAX and WIG serving as larger-market comparators. Using data from January 2012 to December 2025, distributional changes are examined both in relation to volatility and across different parts of the return distribution. The results reveal substantial heterogeneity within the Baltic region. Tallinn exhibits a weak relationship between normalized Wasserstein distance and volatility, closely resembling the behaviour of DAX and WIG. In contrast, Riga and Vilnius show substantially stronger associations, particularly during high-volatility periods. A multivariate analysis including volatility, skewness, and kurtosis leads to the same distinction. Quantile-level analysis further shows that stress-related changes in distributional shape differ considerably across the three Baltic markets. Overall, the results do not support treating Baltic markets as homogeneous with respect to distributional-shape dynamics. Instead, Tallinn displays behaviour closer to the larger reference markets, while Riga and Vilnius exhibit a distinct and more volatility-related pattern.

Such questions were asked:

- Q1.** Are the Baltic stock markets homogeneous with respect to changes in the shape of their normalized return distributions?
- Q2.** To what extent is the change in distributional shape, measured by the normalized W_2 distance, related to conventional measures of volatility, skewness, and kurtosis?
- Q3.** Does the relationship between volatility and distributional-shape change differ between Tallinn, Riga, and Vilnius?
- Q4.** Do Baltic markets exhibit a distributional response to volatility stress that is distinct from that observed in larger European markets represented by the DAX and WIG?
- Q5.** How does the concentration of changes in distributional shape in particular parts of the return distribution differ between the Baltic and larger European markets?

1 BALTIC STOCK MARKETS AND EXISTING LITERATURE

The Baltic equity markets of Estonia, Latvia, and Lithuania constitute a group of relatively small European stock markets with closely connected institutional and economic backgrounds. The national exchanges were established in the 1990s, following the transition of the Baltic economies towards market-based economic systems. The three countries subsequently joined the European Union in 2004 and adopted the euro, with Estonia joining the euro area in 2011, Latvia in 2014, and Lithuania in 2015. Despite this institutional convergence, the three equity markets remain small relative to the major European markets. Gil-Alana et al. (2018) describe the Baltic markets as small markets with low capitalisation and trading volumes.

The three markets are nevertheless closely connected. Brännäs et al. (2012) study the joint dynamics of returns and volatilities in the Tallinn, Riga, and Vilnius markets and find significant cross-market dependencies. In particular, their nonlinear multivariate analysis identifies recursive

relationships between the returns of the three markets, while the volatility of Riga and Vilnius is found to depend on Tallinn. They also report asymmetric effects associated with shocks originating both within the Baltic markets and in the Russian market. These findings provide evidence that geographical proximity and common institutional characteristics do not imply independent market dynamics.

A substantial part of the subsequent literature has focused on the integration of the Baltic markets with larger international markets. Nikkinen, Piljak, and Äijö (2012) [8] examine the relationship between the Baltic markets and the EURO STOXX 50 during the 2008–2009 financial crisis. They find that the Baltic markets were relatively segmented before the crisis but became highly integrated with European markets during the crisis. Their quantile analysis further indicates that the lower tail of Baltic returns was particularly sensitive to movements in the European market during the crisis. This result is particularly relevant for the present study because it suggests that relationships between Baltic and larger markets may change substantially in periods of financial stress and may not be adequately described by unconditional measures of dependence.

Harkmann (2022) extends the analysis of international integration using weekly data from 2005–2015. Her cointegration analysis finds evidence of long-run integration between the Baltic markets and Sweden, together with transmission of shocks from the Swedish market to the Baltic states, while finding less evidence of long-run relationships with the euro-area markets considered in the study.

Liquidity is another important structural feature of these markets. Studies of Baltic stock liquidity find that company characteristics, particularly firm size, are related to liquidity, while earlier research documents the sensitivity of Baltic market liquidity to macroeconomic conditions. These characteristics are relevant when comparing Baltic indices with large developed-market benchmarks, since differences in market size, trading activity, and constituent structure can affect the statistical properties of observed returns.

2 INTRODUCTION

The central object used is the Wasserstein distance W_p between empirical distributions of financial returns. The motivation for this choice is that financial market dynamics are naturally described not only by changes in the first two moments of returns, but also by changes in the shape of their distributions, including asymmetry and tail behaviour. Unlike moment-based measures, the Wasserstein distance compares the distributions themselves and therefore is a natural choice for comparing distributions of points on a metric space $(\mathcal{X}, d)$ [2]. This makes it particularly suitable for studying changes in the distribution of returns over time.

The use of Wasserstein distances in financial applications has received increasing attention. Horváth, Issa and Muguruza [2] formulate market-regime detection directly as a problem on a space of probability measures, using the p -Wasserstein distance between empirical distributions of returns. Their approach is explicitly non-parametric and therefore does not require the return distributions to belong to a prescribed family. More generally, Wasserstein distances provide a mathematically well-established framework for comparing probability distributions and have been extensively studied from both probabilistic and statistical perspectives; see, for example, [10, 9].

2.1 p -Wasserstein Distance

Definition 1. (p -Wasserstein distance) *Let $(\mathcal{X}, d)$ be a separable Radon space and let $\mathcal{P}_p(\mathcal{X})$ denote the set of probability measures having finite p -th moment, where $p \geq 1$. For two probability measures $\mu, \nu \in \mathcal{P}_p(\mathcal{X})$, the p -Wasserstein distance is defined by*

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p d\pi(x, y) \right)^{1/p}, \quad (1)$$

where

$$\Pi(\mu, \nu) := \{\mathbb{P} \in \mathcal{P}(X \times X) : \mathbb{P}(A \times X) = \mu(A), \mathbb{P}(X \times B) = \nu(B)\}$$

denotes the set of all couplings of μ and ν .

The p -Wasserstein distance (14) is the solution to the Kantorovich-type optimal transportation problem between measures μ and ν for the cost function $c(x; y) = d(x; y)^p$. For present applications, existence of an optimal plan $\mathbb{P}^* \in \Pi(\mu; \nu)$ realising the Wasserstein distance between measures $\mu, \nu \in \mathcal{P}(\mathbb{P})$ is guaranteed by continuity of the metric $d(x; y) = |x - y|^p$ and the fact that μ, ν will be empirical measures and thus possess compact support.

The Wasserstein distance is a metric on $\mathcal{P}_p(\mathcal{X})$. [2]

A broader statistical treatment of Wasserstein distances is given by Panaretos and Zemel [9], while a mathematical treatment of optimal transport and Wasserstein spaces can be found in [10].

For the present application, the underlying space is the real line,

$$\mathcal{X} = \mathbb{R},$$

because each observation is a univariate daily log-return. This one-dimensional setting is particularly convenient since the Wasserstein distance admits an explicit representation in terms of quantile functions.

2.2 One-Dimensional Quantile Representation

Definition 2 (Quantile function). Let F_μ and F_ν denote the cumulative distribution functions of μ and ν , respectively, and let

$$F_\mu^{-1}(u) = \inf\{x \in \mathbb{R} : F_\mu(x) > u\}, \quad u \in [0, 1),$$

denote the corresponding generalized quantile function.

Proposition 1. For probability measures on the real line with finite p -th moments, the Wasserstein distance has the particularly simple representation

$$W_p(\mu, \nu) = \left(\int_0^1 |F_\mu^{-1}(u) - F_\nu^{-1}(u)|^p du \right)^{1/p}. \quad (2)$$

Proof. A consequence of the fact that the (unique) optimal transport map pushing μ onto ν is given by $T(x) = (F_\nu^{-1} \circ F_\mu)(x)$, and applying a change of variables. $\square$

This result is fundamental for the present analysis. It shows that the Wasserstein distance compares corresponding quantiles of two distributions. Consequently, the statistic incorporates differences throughout the entire distribution rather than focusing on a particular moment or a selected part of the support. The one-dimensional representation is discussed, in particular, in [10, 9].

For $p = 2$, which is the principal choice in this study,

$$W_2(\mu, \nu) = \left(\int_0^1 |F_\mu^{-1}(u) - F_\nu^{-1}(u)|^2 du \right)^{1/2}. \quad (3)$$

2.3 Empirical Measures

In what follows it is assumed that μ, ν are empirical measures with equal number of atoms $N \in \mathbb{N}$. Let

$$x = (x_1, \dots, x_N), \quad y = (y_1, \dots, y_N)$$

be samples of N returns. Their empirical probability measures are

$$\hat{\mu} = \frac{1}{N} \sum_{j=1}^N \delta_{x_j}, \quad \hat{\nu} = \frac{1}{N} \sum_{j=1}^N \delta_{y_j}, \quad (4)$$

where δ_z denotes the Dirac measure concentrated at z .

Let

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(N)}, \quad y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$$

denote the order statistics of the samples. Then the one-dimensional Wasserstein distance simplifies to

$$W_p(\hat{\mu}, \hat{\nu})^p = \frac{1}{N} \sum_{j=1}^N |x_{(j)} - y_{(j)}|^p. \quad (5)$$

Consequently,

$$W_p(\hat{\mu}, \hat{\nu}) = \left[\frac{1}{N} \sum_{j=1}^N |x_{(j)} - y_{(j)}|^p \right]^{1/p}. \quad (6)$$

This is the empirical representation given in equation (21) of Horvath, Issa and Muguruza [2], following their discussion of Proposition 2.5.

It is particularly important for the present application because it eliminates the need to estimate a probability density function. The comparison is performed directly on the ordered observations.

For $p = 2$,

$$W_2(\hat{\mu}, \hat{\nu}) = \left[\frac{1}{N} \sum_{j=1}^N (x_{(j)} - y_{(j)})^2 \right]^{1/2}. \quad (7)$$

An important computational consequence is that the observations only need to be sorted. [2] note that, for empirical measures with N atoms, the computation is $O(N \log N)$ when sorting is required and becomes linear once the observations have already been sorted. This makes the measure particularly suitable for repeated calculations over rolling windows.

In the present study, each rolling window contains the same number of observations. This is important because equation (5) then compares equally weighted empirical distributions with the same number of atoms. The Wasserstein distance is therefore computed directly from the ordered returns in the two windows.

2.4 Shape Standardization

(The following scale decomposition of W_2 is established in papers by Lipp & Vermeesch (2023) [7] and Irpino, A. and Romano, E. (2007) [3])

Assume one-dimensional case. Let Q_μ and Q_ν denote the quantile functions of two probability distributions μ and ν , respectively. Then

$$W_2(\mu, \nu)^2 = \int_0^1 (Q_\mu(u) - Q_\nu(u))^2 du. \quad (8)$$

Let the means and standard deviations of the two distributions be $m_\mu, m_\nu, \sigma_\mu, \sigma_\nu$. Define the standardized quantile functions

$$Z_\mu(u) = \frac{Q_\mu(u) - m_\mu}{\sigma_\mu}, \quad Z_\nu(u) = \frac{Q_\nu(u) - m_\nu}{\sigma_\nu}. \quad (9)$$

Consequently,

$$Q_\mu(u) = m_\mu + \sigma_\mu Z_\mu(u), \quad Q_\nu(u) = m_\nu + \sigma_\nu Z_\nu(u). \quad (10)$$

Since the mean and variance of a distribution can be expressed through its quantile function,

$$m_\mu = \int_0^1 Q_\mu(u) du, \quad \sigma_\mu^2 = \int_0^1 (Q_\mu(u) - m_\mu)^2 du, \quad (11)$$

and analogously for ν . It follows from (9) that

$$\int_0^1 Z_\mu(u) du = 0, \quad \int_0^1 Z_\mu(u)^2 du = 1, \quad (12)$$

Substituting (10) into (8) gives

$$W_2(\mu, \nu)^2 = \int_0^1 [(m_\mu - m_\nu) + \sigma_\mu Z_\mu(u) - \sigma_\nu Z_\nu(u)]^2 du. \quad (13)$$

Expanding the square yields

$$\begin{aligned} W_2(\mu, \nu)^2 &= (m_\mu - m_\nu)^2 + \sigma_\mu^2 \int_0^1 Z_\mu(u)^2 du + \sigma_\nu^2 \int_0^1 Z_\nu(u)^2 du \\ &\quad - 2\sigma_\mu \sigma_\nu \int_0^1 Z_\mu(u) Z_\nu(u) du \\ &\quad + 2(m_\mu - m_\nu) \int_0^1 [\sigma_\mu Z_\mu(u) - \sigma_\nu Z_\nu(u)] du. \end{aligned} \quad (14)$$

The last integral vanishes because of (12). Moreover,

$$\int_0^1 Z_\mu(u)^2 du = \int_0^1 Z_\nu(u)^2 du = 1.$$

Therefore,

$$W_2(\mu, \nu)^2 = (m_\mu - m_\nu)^2 + \sigma_\mu^2 + \sigma_\nu^2 - 2\sigma_\mu \sigma_\nu \int_0^1 Z_\mu(u) Z_\nu(u) du. \quad (15)$$

Define

$$\rho_Q = \int_0^1 Z_\mu(u) Z_\nu(u) du. \quad (16)$$

Since both standardized quantile functions have zero mean and unit variance, ρ_Q is their Pearson correlation. Hence

$$W_2(\mu, \nu)^2 = (m_\mu - m_\nu)^2 + (\sigma_\mu - \sigma_\nu)^2 + 2\sigma_\mu \sigma_\nu (1 - \rho_Q). \quad (17)$$

Equation (17) separates the squared Wasserstein distance into three components. The first, $(m_\mu - m_\nu)^2$, represents a difference in location; the second, $(\sigma_\mu - \sigma_\nu)^2$, represents a difference in scale; and the final term, $2\sigma_\mu \sigma_\nu (1 - \rho_Q)$, represents the difference between the standardized shapes of the two distributions. This location-scale-shape interpretation of the one-dimensional Wasserstein distance is discussed in the distributional-data literature; see, for example, [3, 4] and the more recent treatment of location-scale-shape models in Wasserstein geometry [1].

This decomposition is important for the present application because the raw Wasserstein distance does not isolate changes in distributional shape. In particular, two distributions may have exactly the same standardized shape while having different standard deviations. Thus, a change in volatility alone is sufficient to generate a non-zero Wasserstein distance, even when the underlying distributional shape has remained unchanged.

To accomplish this, each rolling window of returns is standardized independently. Let

$$R_t = \{r_{t-n+1}, \dots, r_t\}$$

be the set of returns in a rolling window of length n . Its sample mean and standard deviation are denoted by $\bar{r}_t$ and s_t , respectively. Each observation in the window is transformed according to

$$z_{t-i} = \frac{r_{t-i} - \bar{r}_t}{s_t}, \quad i = 0, \dots, n-1. \quad (18)$$

Let $\hat{\mu}_t^z$ denote the empirical distribution of these standardized observations. By construction, $\hat{m}_t^z = 0, \hat{\sigma}_t^z = 1$.

Consider two consecutive standardized windows, $\hat{\mu}_t^z$ and $\hat{\mu}_{t-1}^z$. Applying the decomposition (17) gives

$$\begin{aligned} W_2^{\text{norm}}(t)^2 &= (0 - 0)^2 + (1 - 1)^2 + 2(1)(1)(1 - \rho_{Q,t}) \\ &= 2(1 - \rho_{Q,t}), \end{aligned} \quad (19)$$

where

$$W_2^{\text{norm}}(t) := W_2(\hat{\mu}_t^z(t), \hat{\mu}_t^z(t-1)). \quad (20)$$

Thus, standardization removes the location and scale components of the Wasserstein distance exactly. The resulting statistic measures differences between the standardized distributions. In this sense, W_2^{norm} can be interpreted as a measure of *distributional shape change after location-scale normalization*.

The term ‘shape’ should therefore be understood in the affine sense. Two distributions that differ only through a transformation $x \mapsto a + bx, b > 0$, become identical after standardization and consequently have $W_2^{\text{norm}} = 0$.

On the other hand, differences that cannot be removed by such a location-scale transformation remain visible in the normalized Wasserstein distance. These may include changes in asymmetry, tail thickness, concentration and other features of the standardized distribution.

2.5 Application to Rolling Market Comparisons

The empirical analysis focuses on the normalized 2-Wasserstein distance. As shown in Section 2.4, standardizing each rolling return distribution to zero mean and unit variance removes the location and scale components of W_2 . Consequently,

$$W_2^{\text{norm}}(t)^2 = 2(1 - \rho_Q(t)), \quad (21)$$

where $\rho_Q(t)$ is the correlation between the standardized quantile functions. Thus, W_2^{norm} measures changes in the distribution after removing changes in its mean and overall scale.

This normalization is particularly important for comparing markets with different volatility levels and market sizes. The normalized distance allows the analysis to focus on changes in the standardized distribution, including changes in skewness, kurtosis, and tail behaviour instead of differences in scale.

2.6 Quantile-Level Interpretation

An additional advantage of the one-dimensional Wasserstein formulation is that it permits the contribution of different parts of the distribution to be examined directly. From equation (3),

$$W_2(\mu, \nu)^2 = \int_0^1 [F_\mu^{-1}(u) - F_\nu^{-1}(u)]^2 du. \quad (22)$$

thus other than treating the distribution as an indivisible object, one can examine $[F_\mu^{-1}(u) - F_\nu^{-1}(u)]^2$ as a function of the probability level u . This makes it possible to investigate whether distributional changes are concentrated in the centre, shoulders or tails of the return distribution.

For the empirical analysis, the interval $([0,1])$ is divided into $(K \in \mathbb{N})$ quantile intervals,

$$0 = u_0 < u_1 < \dots < u_K = 1.$$

The contribution of the (k) -th quantile interval is therefore

$$\int_{u_{k-1}}^{u_k} |Q_\mu(u) - Q_\nu(u)|^2 du, \quad (23)$$

This representation allows the source of changes in W_2^{norm} to be examined across different parts of the return distribution. In particular, the analysis can distinguish changes concentrated in the centre, shoulders, or tails, providing information that cannot be obtained from volatility alone.

3 METHODOLOGY AND NUMERICAL RESULTS

The analysis uses daily closing prices of the Tallinn (OMXT), Riga (OMXR), and Vilnius (OMXV) stock indices, together with the DAX and WIG indices as larger-market comparators. (For note on Index Selection, see Section 4.1.) All data were obtained from Stooq.com and cover the period from January 2012 to December 2025. Logarithmic returns are calculated from consecutive daily closing prices. For each index, 30-observation rolling windows are used to construct empirical return distributions, which are subsequently standardized to zero mean and unit variance before calculating the normalized W_2 distance.

3.1 Distributional Shape under Market Stress

The normalized W_2 distance is calculated between consecutive rolling distributions according to the quantile representation introduced in Section 2.3. This removes differences in location and scale and therefore allows the analysis to focus on changes in the standardized shape of the return distribution. Volatility is calculated for the same rolling windows and is used as a measure of the scale of return fluctuations.

To examine whether distributional changes differ between ordinary and high-volatility periods, stress periods are defined as observations for which rolling volatility exceeds its 90th percentile. The analysis compares the relationship between W_2^{norm} and volatility across the two regimes and calculates

$$\mathbb{E}[W_2^{\text{norm}} \mid \text{normal}], \quad \mathbb{E}[W_2^{\text{norm}} \mid \text{stress}], \quad (24)$$

as well as their ratio,

$$R_q = \frac{\mathbb{E}[W_2^{\text{norm}} \mid \text{stress}]}{\mathbb{E}[W_2^{\text{norm}} \mid \text{normal}]}. \quad (25)$$

The obtained results are presented beneath in Figure 1 and Table 1.

Index	$\mathbb{E}[W_2^{\text{norm}} \mid \text{normal}]$	$\mathbb{E}[W_2^{\text{norm}} \mid \text{stress}]$	R
DAX	0.2840479584	0.2869445974	1.0101977109
WIG	0.2717970110	0.2719656112	1.0006203166
OMXT	0.3206596744	0.3520204891	1.0978009309
OMXR	0.3465885366	0.5092601718	1.4693508814
OMXV	0.3119960688	0.4385681582	1.4056848851

Table 1: Expected value of W_2 for stress and normal situations with indicator R computed.

The results show substantial differences between the markets in the behaviour of normalized Wasserstein distance during high-volatility periods. For the Riga and Vilnius indices, W_2^{norm} exhibits a strong relationship with volatility during stress periods. In contrast, this relationship is considerably weaker for Tallinn, DAX, and WIG. Thus, increased volatility in Riga and Vilnius is more frequently accompanied by changes in the standardized shape of the return distribution.

The unconditional levels of normalized Wasserstein distance also differ across the markets. The expected values during normal periods for Tallinn, Riga, and Vilnius are relatively similar and are higher than those observed for DAX and WIG. This indicates that the Baltic markets generally exhibit greater distributional variation in ordinary periods after controlling for location and scale.

However, the stress-to-normal ratio R_q provides a different comparison. The value obtained for Tallinn is substantially closer to those observed for DAX and WIG than to the corresponding values for Riga and Vilnius. Hence, although Tallinn has a similar level of ordinary-period distributional variation to the other Baltic markets, its relative change in distributional shape between normal and high-volatility periods is more similar to that observed in the larger comparison markets.

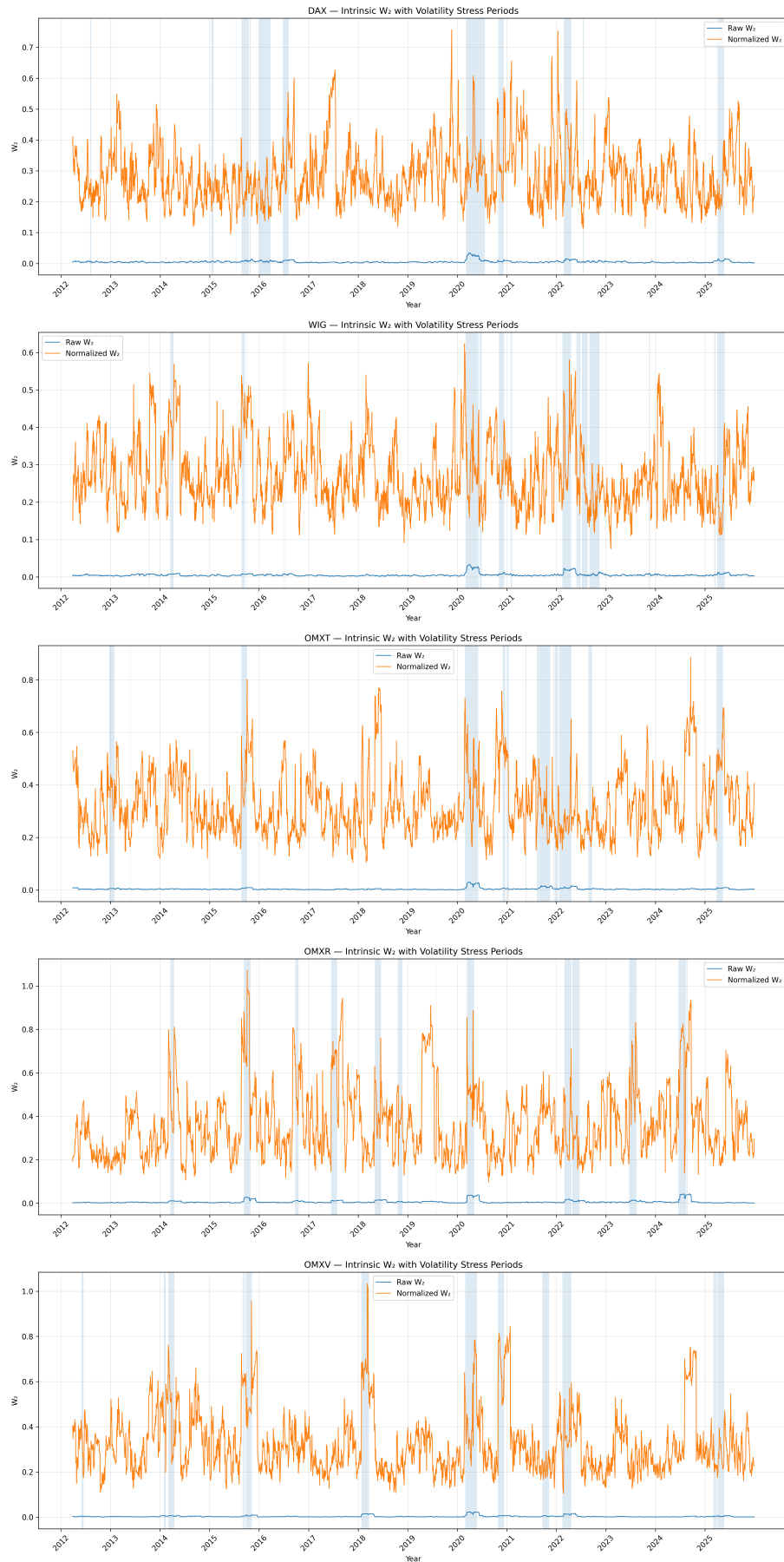


Figure 1: W_2 both raw and normalized calculated for DAX, WIG, OMXT, OMXR, OMXV. Stress periods (volatility over 90th percentile) are marked with blue.

3.2 Distributional Shape and Conventional Moments

Second study is made to examine, whether correlation between W_2^{norm} and volatility is stronger for Vinius and Riga than Tallinn, DAX and WIG only in stress periods or in overall distribution.

For each index, the linear correlation is computed between W_2^{norm} and

1. rolling volatility (σ_t) using Pearson, Spearman, Distance Correlation and Ordinary Least Sequences (OLS) with coefficients of determination r, ρ, r_d, R_1^2 respectively (Figure 2. and Table 2.); and
2. rolling volatility, skewness and kurtosis ($\sigma(t), S(t), K(t)$), using the following model

$$W_2^{\text{norm}}(t) = \beta_0 + \beta_1\sigma(t) + \beta_2S(t) + \beta_3K(t) + \varepsilon(t),$$

with adjusted coefficient of determination R_2^2 (Table 3.).

Index	Pearson	Spearman	Dist. corr.	R_1^2
DAX	-0.01242748	-0.07592692	0.09811600	0.00015444
WIG	-0.00566377	-0.03894444	0.06207654	0.00003208
OMXT	0.02134310	-0.03327477	0.09368649	0.00045553
OMXR	0.34567321	0.23397434	0.34764560	0.11948997
OMXV	0.29818900	0.30676518	0.32968138	0.08891668

Table 2: Coefficients of determination for fitting W_2^{norm} to $\sigma(t)$.

Based on data in Table 2. a clear distinction can be made between DAX, WIG, OMXT (Group 1) and OMXR, OMXV (Group 2). G2 appears to have much better linear correlation between W_2^{norm} and volatility, while all presented coefficient of determination for OMXT tend to be close to zero, similarly to those of DAX and WIG.

This shows that shape of market distribution of Tallinn tends to have distributional-shape changes less correlated to volatility, hence resembling changes of shape associated with larger European markets.

There is evidence ([8]) that Baltic markets tend to become substantially more integrated during the stress periods, yet coefficient of determination for fits in normal and stress condition were measured to have much similar values. Hence, the results provide no evidence that W_2^{norm} and volatility are more correlated during stress periods.

Overall, the results reinforce the lack of distribution-shape homogeneity across the Baltic markets. In particular, Tallinn behaves more similarly to the larger DAX and WIG markets than to its Baltic counterparts, while Riga and Vilnius display a stronger volatility–distribution relationship.

Index	R_2^2
DAX	0.107705
WIG	0.148545
OMXT	0.114425
OMXR	0.321051
OMXV	0.291032

Table 3: Coefficients of determination for fitting W_2^{norm} to $\sigma(t), S(t), K(t)$.

Including skewness and kurtosis in the linear model strengthens the previous finding (Table 3.). The resulting R_2^2 remains relatively low for DAX, WIG, and OMXT, at 0.108, 0.149, and 0.114, respectively. In contrast, OMXR and OMXV reach substantially higher values of 0.321 and 0.291.

Thus, even when volatility is supplemented by the first two higher-order distributional moments, W_2 remains largely unexplained for Tallinn, similarly to DAX and WIG. The stronger explanatory power for Riga and Vilnius further supports the conclusion that the relationship between Wasserstein distributional changes and conventional distributional characteristics is not homogeneous across the Baltic markets.

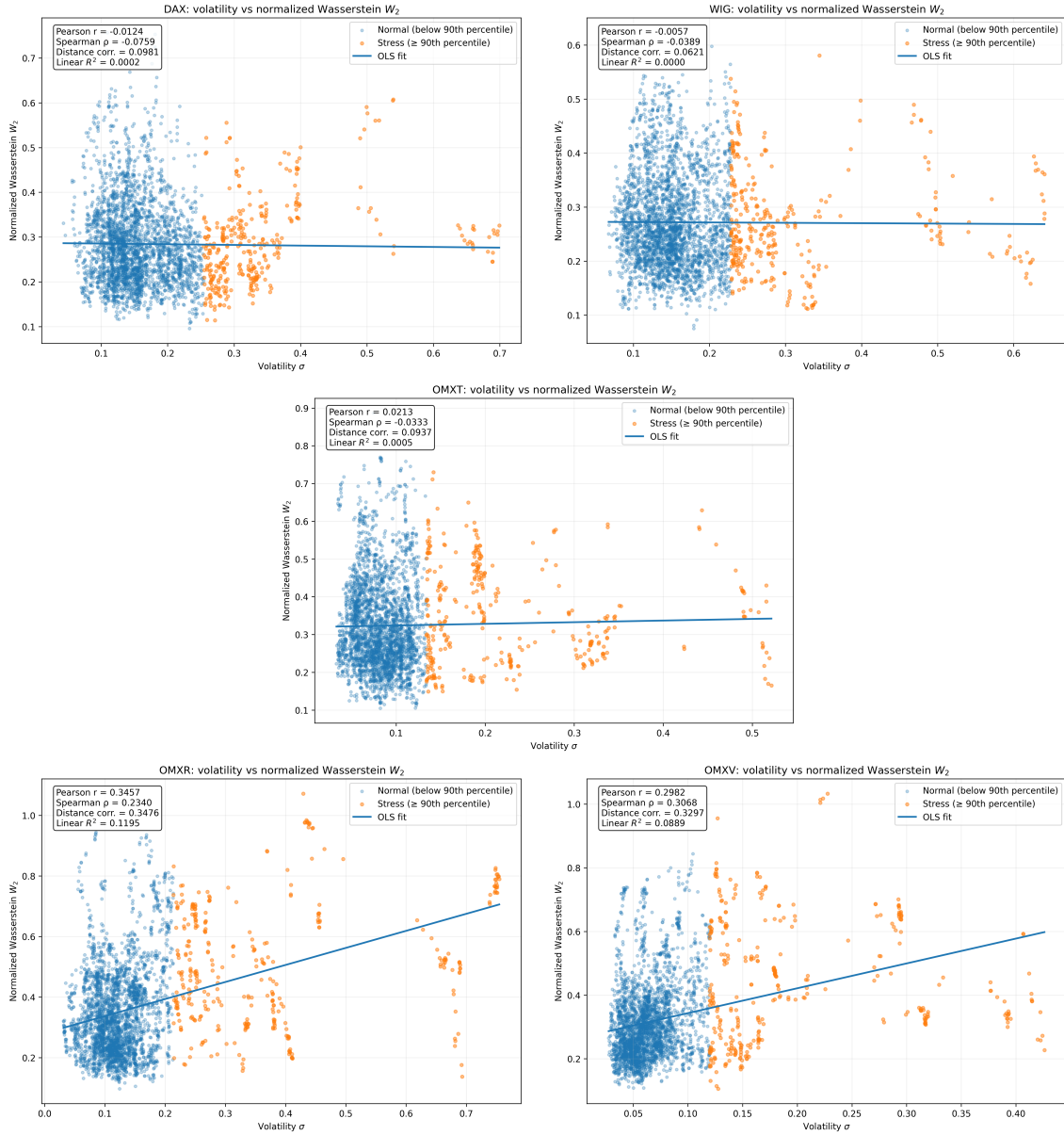


Figure 2: Volatility vs W_2^{norm} divided into normal and stress regimes, with Pearson, Spearman, Distance and OLS correlations; (OLS fit added only illustratively).

3.3 Quantile-Level Analysis

Last study is made to examine in which quantiles of the distribution does W_2^{norm} differ for all the indices. The rolling distributions are, thus, split into quantiles:

$$(0, 0.05), (0.05 - 0.25), (0.25 - 0.75), (0.75 - 0.95), (0.95, 1),$$

than for each quantile and stress regime (depending on volatility -- defined as before) an expected value is calculated according to theory in Section 2.6. (Table 4.)

Since numbers of observations for both tails are small (1-2 observations), due to small window length the estimations of expected value of W_2^{norm} might not be accurate. To address this issue, we

define

$$R_q = \frac{\mathbb{E}[W_2^{\text{norm}} \mid \text{stress}]}{\mathbb{E}[W_2^{\text{norm}} \mid \text{normal}]}, \quad (26)$$

then use bootstrap confidence intervals. To adress the issue that rolling windows are short-range temporaly dependant on each other, a Moving-Block Bootstrap (MBB) is used to resample consecutive blocks rather than individual observations (Figure 3 and Table 4.). For additional theory see Künsch (1989) [5] and Lahiri (2003) [6].

Additional robustness test were conducted using windows of 60 and 120 observations, yet as the results are much similar they are not included.

Index	Quantile	$\mathbb{E}[W_2^{\text{norm}} \mid \text{normal}]$	$\mathbb{E}[W_2^{\text{norm}} \mid \text{stress}]$	R	Error interval
DAX	0-5%	0.047589944	0.033861036	0.7115	[0.4079, 1.0905]
	5-25%	0.061971622	0.055670555	0.8983	[0.6823, 1.0573]
	25-75%	0.059144825	0.068785458	1.163	[1.0453, 1.3274]
	75-95%	0.05428795	0.046758465	0.8613	[0.6952, 1.0352]
	95-100%	0.041964386	0.043270226	1.0311	[0.7291, 1.4495]
WIG	0-5%	0.039440781	0.031970383	0.8106	[0.2536, 1.3811]
	5-25%	0.055994332	0.060401402	1.0787	[0.8360, 1.4209]
	25-75%	0.062493761	0.06674179	1.068	[0.9534, 1.2516]
	75-95%	0.054311661	0.050502517	0.9299	[0.8004, 1.0708]
	95-100%	0.039526813	0.035498199	0.8981	[0.6893, 1.1960]
OMXT	0-5%	0.051053224	0.044248386	0.8667	[0.5181, 1.3572]
	5-25%	0.056390089	0.076789988	1.3618	[1.0844, 1.7966]
	25-75%	0.055905189	0.059765592	1.0691	[0.9069, 1.2808]
	75-95%	0.053553743	0.055217937	1.0311	[0.8387, 1.2615]
	95-100%	0.053491989	0.03951903	0.7388	[0.5219, 1.1180]
OMXR	0-5%	0.047044062	0.078517415	1.669	[0.9552, 2.3380]
	5-25%	0.054709441	0.06935446	1.2677	[1.0415, 1.5076]
	25-75%	0.05649753	0.045943275	0.8132	[0.7152, 0.9253]
	75-95%	0.052099212	0.062968633	1.2086	[0.9616, 1.5344]
	95-100%	0.054836431	0.086951572	1.5857	[1.1004, 1.9549]
OMXV	0-5%	0.045240662	0.079095186	1.7483	[1.1586, 2.7123]
	5-25%	0.055363618	0.084957332	1.5345	[1.2830, 1.9395]
	25-75%	0.059391333	0.056091607	0.9444	[0.8130, 1.0852]
	75-95%	0.054038698	0.049145049	0.9094	[0.6706, 1.1276]
	95-100%	0.051906126	0.051865282	0.9992	[0.7153, 1.4781]

Table 4: Expeted value of W_2^{norm} in each quantile with R computed with MBB and Error intervals for each index.

The results indicate that the Baltic markets do not form a homogeneous group with respect to the response of W_2^{norm} to volatility stress. Tallinn (OMXT) is relatively close to DAX and WIG: its R remains around unity for the central and upper quantiles and falls below one in the most extreme 95% – 100% quantile. Riga and Vilnius behave differently, showing markedly elevated R in the lower and moderately high volatility regimes.

The distinction becomes especially clear in the extreme quantiles. Riga shows $R = 1.586$ in the 95% – 100% regime, while Vilnius returns to approximately unity ($R = 0.999$); Tallinn instead falls to $R = 0.739$. Thus, even within the Baltic region, stress does not induce a common transformation of the return-distribution shape.

Compared with DAX and WIG, the Baltic markets exhibit larger and more heterogeneous deviations of R from unity, particularly in the lower-volatility and intermediate regimes. DAX and WIG remain relatively close to $R = 1$ across most quantiles, whereas Riga and Vilnius display stronger stress-related increases in W_2^{norm} . Together with the previous correlation analysis, this supports the conclusion that Baltic markets cannot be characterised by a single distributional

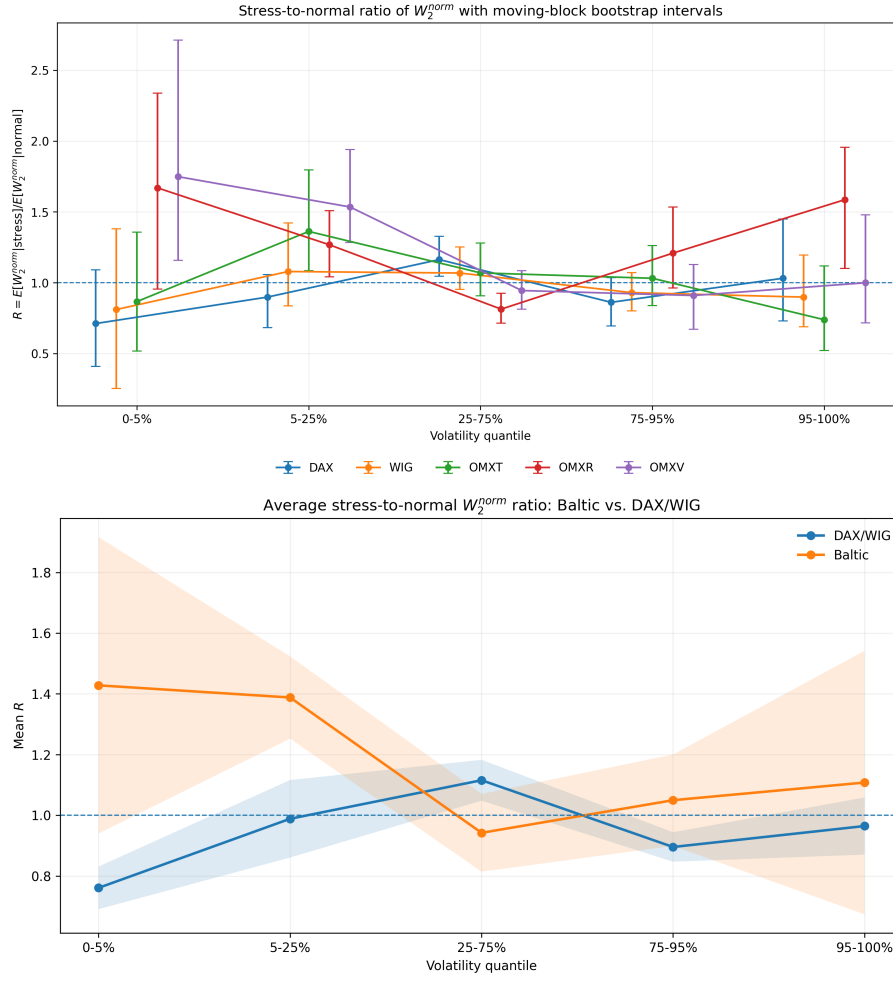


Figure 3: Ratio of expected value of W_2^{norm} in stress to normal in each quantile of the rolling distributions.

response to volatility: Tallinn is closer to the behaviour of the larger reference markets, while Riga and Vilnius exhibit substantially different distribution-shape dynamics.

4 CONCLUSION

This project examined whether the Baltic stock markets can be regarded as homogeneous with respect to changes in the shape of their return distributions. Using the normalized 2-Wasserstein distance, the analysis shows that they cannot. Although the three markets share a common regional setting, their distributional responses to volatility differ substantially. Tallinn exhibits a very weak relationship between W_2^{norm} and volatility, while Riga and Vilnius display considerably stronger dependence. The same distinction remains when skewness and kurtosis are included, with the explanatory power for Tallinn remaining close to that of DAX and WIG, and substantially higher values being found for Riga and Vilnius.

The stress analysis reinforces this distinction. The stress-to-normal ratio R_q remains relatively close to unity for DAX, WIG, and Tallinn across most quantile ranges, whereas Riga and Vilnius show pronounced increases in W_2^{norm} in several volatility regimes. These results indicate that high volatility does not induce a common transformation of distributional shape across the Baltic markets.

The comparison with DAX and WIG provides an additional perspective. Tallinn is consistently closer to the larger reference markets in the relationship between volatility and distributional-shape changes, despite being geographically and economically part of the Baltic group. Riga and Vilnius, on the other hand, show a stronger association between volatility and changes in the standardized return distribution. This distinction suggests that geographical proximity and common regional classification are not sufficient to imply homogeneous distributional dynamics. Moreover, the finding that the relationship between W_2^{norm} and volatility does not become systematically stronger in stress periods indicates that increased market integration during crises does not necessarily translate into a common volatility–distributional-shape mechanism.

4.1 Index selection and comparability

The choice of indices is subjected to an important data limitation. OMXT, OMXR, and OMXV were selected as the most appropriate broad national-market indices available for a consistent analysis of the Tallinn, Riga, and Vilnius markets over the study period. Despite differences in absolute market size, the three indices have broadly similar structural characteristics. In particular, their capitalisation is similarly concentrated among a relatively small number of large companies, making their composition more comparable with each other than their absolute market sizes might suggest. Furthermore, OMXT and OMXV are relatively similar in terms of total market capitalisation, whereas the Riga market is substantially smaller.

The Baltic markets are nevertheless substantially smaller and structurally different from the Polish and German markets represented by WIG and DAX. They also exhibit higher volatility, which is consistent with characteristics commonly associated with smaller and less liquid equity markets. Consequently, the purpose of including DAX and WIG is not to construct perfectly matched indices, but to provide larger-market reference points against which the distributional dynamics of the Baltic markets can be interpreted.

This difference in market structure is also one reason why the normalized Wasserstein distance is particularly useful in the present setting. Since standardization removes the location and scale components of W_2 , comparisons are less directly driven by the different volatility levels of the underlying markets. The comparison therefore focuses on differences in the shape of the standardized return distributions rather than simply differences in the absolute scale of returns.

These considerations make the three OMX indices suitable for examining the central question of the study: whether markets that are structurally comparable within the Baltic region nevertheless exhibit homogeneous distributional-shape dynamics. The results indicate that they do not. While Riga and Vilnius display broadly similar behaviour, Tallinn repeatedly follows a pattern closer to the larger DAX and WIG markets.

References

- [1] Ayumu Fukushi, Yoshinori Nakanishi-Ohno, and Takeru Matsuda. “Flatness of location-scale-shape models under the Wasserstein metric”. In: *Information Geometry* (2026). DOI: 10.1007/s41884-026-00206-6.
- [2] Blanka Horváth, Zacharia Issa, and Aitor Muguruza. “Clustering market regimes using the Wasserstein distance”. In: *Journal of Computational Finance* 28.1 (2024), pp. 1–39. DOI: 10.21314/JCF.2024.005.
- [3] Antonio Irpino and Elvira Romano. “Optimal histogram representation of large data sets: Fisher vs piecewise linear approximation”. In: *European Conference on Principles of Data Mining and Knowledge Discovery*. 2007, pp. 99–110.
- [4] Antonio Irpino and Rosanna Verde. “Basic statistics for distributional symbolic variables: a new metric-based approach”. In: *Advances in Data Analysis and Classification* 9 (2015), pp. 279–297. DOI: 10.1007/s11634-014-0176-4.
- [5] Hans R. Künsch. “The Jackknife and the Bootstrap for General Stationary Observations”. In: *The Annals of Statistics* 17.3 (1989), pp. 1217–1241. DOI: 10.1214/aos/1176347265.
- [6] Soumendra Nath Lahiri. *Resampling Methods for Dependent Data*. New York: Springer, 2003. DOI: 10.1007/978-1-4757-3803-2.
- [7] Adam G. Lipp and Pieter Vermeesch. “Short communication: The Wasserstein distance as a dissimilarity metric for comparing detrital age spectra, and other geological distributions”. In: *Geochronology* 5 (2023), pp. 263–271. DOI: 10.5194/gchron-5-263-2023.
- [8] Jussi Nikkinen, Viktor Piljak, and Janne Äijö. “Baltic stock markets and the financial crisis of 2008–2009”. In: *Research in International Business and Finance* 26.3 (2012), pp. 398–409. DOI: 10.1016/j.ribaf.2012.03.003.
- [9] Victor M. Panaretos and Yoav Zemel. “Statistical Aspects of Wasserstein Distances”. In: *Annual Review of Statistics and Its Application* 6 (2019), pp. 405–431. DOI: 10.1146/annurev-statistics-030718-104938.
- [10] Filippo Santambrogio. *Optimal Transport for Applied Mathematicians: Calculus of Variations, PDEs, and Modeling*. Birkhäuser, 2015. DOI: 10.1007/978-3-319-20828-2.